

Problem

Let

$$F(s) = \frac{18(s+1)}{(s+2)(s+3)}$$

(a) Use the initial and final value theorems to find $f(0)$ and $f(\infty)$.

(b) Verify your answer in part (a) by finding $f(t)$, using partial fractions.

Step-by-step solution

Step 1 of 8

(a)

Determine the value of $f(0)$ using initial value theorem.

$$f(0) = \lim_{s \rightarrow \infty} s F(s)$$

Substitute $\frac{18(s+1)}{(s+2)(s+3)}$ for $F(s)$ in the equation.

$$\begin{aligned} f(0) &= \lim_{s \rightarrow \infty} s \left(\frac{18(s+1)}{(s+2)(s+3)} \right) \\ &= \lim_{s \rightarrow \infty} s \left(\frac{18s \left(1 + \frac{1}{s} \right)}{s^2 \left(1 + \frac{2}{s} \right) \left(1 + \frac{3}{s} \right)} \right) \\ &= \lim_{s \rightarrow \infty} \left(\frac{18s^2 \left(1 + \frac{1}{s} \right)}{s^2 \left(1 + \frac{2}{s} \right) \left(1 + \frac{3}{s} \right)} \right) \end{aligned}$$

Further simplification as follows,

$$\begin{aligned} f(0) &= \lim_{s \rightarrow \infty} \left(\frac{18 \left(1 + \frac{1}{s} \right)}{\left(1 + \frac{2}{s} \right) \left(1 + \frac{3}{s} \right)} \right) \\ &= \frac{18 \left(1 + \frac{1}{\infty} \right)}{\left(1 + \frac{2}{\infty} \right) \left(1 + \frac{3}{\infty} \right)} \\ &= \frac{18(1+0)}{(1+0)(1+0)} \\ &= 18 \end{aligned}$$

Thus, the initial value of transfer function is, $f(0) = 18$.

Step 2 of 8

Determine the value of $f(\infty)$ using final value theorem.

$$f(\infty) = \lim_{s \rightarrow 0} s F(s)$$

Substitute $\frac{18(s+1)}{(s+2)(s+3)}$ for $F(s)$ in the equation.

$$\begin{aligned} f(\infty) &= \lim_{s \rightarrow 0} s \left(\frac{18(s+1)}{(s+2)(s+3)} \right) \\ &= 0 \left(\frac{18(0+1)}{(0+2)(0+3)} \right) \\ &= 0 \end{aligned}$$

Thus, the final values of transfer function is, $\boxed{f(\infty) = 0}$.

Step 3 of 8

(b)

Consider the following transfer function:

$$F(s) = \frac{18(s+1)}{(s+2)(s+3)}$$

Simplify the transfer function, $F(s)$ using partial fraction method.

$$\frac{18(s+1)}{(s+2)(s+3)} = \frac{A}{(s+2)} + \frac{B}{(s+3)} \dots\dots (1)$$

Here,

A and B are the constants.

Step 4 of 8

Calculate the constant A using residue theorem.

$$\begin{aligned} A &= \left[(s+2) F(s) \right]_{s=-2} \\ &= \left[(s+2) \frac{18(s+1)}{(s+2)(s+3)} \right]_{s=-2} \\ &= \left[\frac{18(s+1)}{(s+3)} \right]_{s=-2} \\ &= \frac{18(-2+1)}{(-2+3)} \\ &= -18 \end{aligned}$$

Step 5 of 8

Calculate the constant B using residue theorem.

$$\begin{aligned}
 B &= \left[(s+3)F(s) \right]_{s=-3} \\
 &= \left[(s+3) \frac{18(s+1)}{(s+2)(s+3)} \right]_{s=-3} \\
 &= \left[\frac{18(s+1)}{(s+2)} \right]_{s=-3} \\
 &= \frac{18(-3+1)}{(-3+2)} \\
 &= 36
 \end{aligned}$$

Step 6 of 8

Recall equation (1),

$$\begin{aligned}
 \frac{18(s+1)}{(s+2)(s+3)} &= \frac{A}{(s+2)} + \frac{B}{(s+3)} \\
 F(s) &= \frac{A}{(s+2)} + \frac{B}{(s+3)}
 \end{aligned}$$

Substitute -18 for A and 36 for B in the equation.

$$F(s) = \frac{-18}{(s+2)} + \frac{36}{(s+3)}$$

Apply inverse transform on both sides,

$$L^{-1}\{F(s)\} = -L^{-1}\left\{\frac{18}{s+2}\right\} + L^{-1}\left\{\frac{36}{s+3}\right\}$$

It is known that,

$$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$$

Therefore, the expression for $f(t)$ is,

$$f(t) = -18e^{-2t} + 36e^{-3t} \dots\dots (2)$$

Step 7 of 8

Determine the initial value of transfer function.

Recall equation (2),

$$f(t) = -18e^{-2t} + 36e^{-3t}$$

Substitute 0 for t in the equation.

$$\begin{aligned}
 f(0) &= -18e^{-2(0)} + 36e^{-3(0)} \\
 &= -18(1) + 36(1) \\
 &= 18
 \end{aligned}$$

Thus, the initial value of transfer function is, $f(0) = 18$.

Step 8 of 8

Determine the final value of transfer function.

Recall equation (2),

$$f(t) = -18e^{-2t} + 36e^{-3t}$$

Substitute ∞ for t in the equation.

$$\begin{aligned} f(\infty) &= -18e^{-2(\infty)} + 36e^{-3(\infty)} \\ &= -18(0) + 36(0) \\ &= 0 \end{aligned}$$

Thus, the final values of transfer function is, $f(\infty) = 0$.